

Classification of Atrial Fibrillation Burden Tiers in ECG Signals Using Hybrid 1D

CNN-LSTM Models

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Chapter 1

INTRODUCTION

This chapter outlines the study's structure. It begins with the clinical context of atrial fibrillation (AF) and its associated health risks, followed by an analysis of current gaps in diagnosing its temporal burden. It then presents the machine learning approach used to address these limitations and introduces the core research questions and objectives. Finally, the section details the significance, scope, and technical boundaries of the research.

1.1. Context of the Study

Atrial fibrillation (AF) is a major cardiovascular condition that increases risks for stroke and heart failure [1, 6]. In computational cardiology, automating the temporal burden tier classification from electrocardiogram (ECG) signals is challenging because physiological signals are non-stationary, with highly variable R-R intervals and missing P-waves. Raw ECG data is also corrupted by baseline wander due to respiration, 50/60 Hz power-line interference, and electromyographic (EMG) artifacts from muscle movement [3]. A computational model must filter these noise types before extracting features.

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To improve upon shallow machine learning algorithms, this study uses a hybrid 1D CNN-LSTM architecture to classify AF burden tiers in ECG Signals, jointly modeling the signal's spatial and temporal dimensions (rationale detailed in **1.5**).

Currently, the clinical standard for classifying ECG results relies on manual interpretation of 12-lead ECGs or Holter monitor recordings by cardiologists using European Society of Cardiology (ESC) guidelines [2]. Specialists visually judge burden tiers based on arrhythmic episode frequency, ventricular rate, and subjective symptom scores, such as the EHRA score. However, this manual approach is time-intensive, does not scale for continuous remote monitoring, and is susceptible to inter-observer variability [4].

1.2. The Problem, Gap, or Opportunity

Although deep learning models detect cardiac issues more effectively than older statistical equations, the existing literature often emphasizes peak performance metrics over theoretical justification for specific network architectures based on ECG signal characteristics [5]. This makes the code uninterpretable and limits clinical trust. Furthermore, current engineering work relies too much on global accuracy. Accuracy is misleading when testing on real-world patient data, where normal heartbeats vastly outnumber abnormal ones.[1] Medical tools require metrics such as precision, recall, and F1 Scores to track dangerous missed detections and prevent false alarms. This study addresses these gaps by classifying cardiac anomalies into distinct temporal burden tiers (Class 0: Sinus Rhythm, Class 1: Micro-Burden /

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Rare Paroxysm, Class 2: Intermediate Burden / Active Paroxysm, and Class 3: High Burden / Persistent AFib) and establishing a transparent evaluation pipeline aligned with industry-standard testing criteria. Furthermore, single ECG scans cannot determine longitudinal risk, so deep learning classifications must be integrated with relational database tracking and clinical risk scales (such as the CHA₂DS₂-VASc stroke risk score) to form a functional Clinical Decision Support System (CDSS).

1.3. Research Questions

- How effective is a hybrid 1D CNN-LSTM model in classifying the burden tiers of ECG signals compared to existing single-architecture methods?
- Which evaluation metrics (accuracy, precision, recall, F1-score, and ROC-AUC) best reflect the model's clinical reliability and performance?

1.4. Research Objectives

The primary goal of this study is to develop and evaluate a 1D CNN-LSTM model for classifying ECG signal burden tiers by leveraging both the spatial and temporal properties of the waveform.

In particular, this research aims to:

- Develop a CNN-LSTM model and benchmark its performance against existing deep learning baselines.

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- Evaluate clinical reliability performance metrics tailored for diagnostic workflows:
 - **Recall (Sensitivity):** To measure the model's capacity to correctly identify positive atrial fibrillation cases, ensuring critical cardiac risks are not missed.
 - **Precision:** To assess positive prediction accuracy, minimize false alarms that trigger clinical fatigue or unnecessary treatment.
 - **F1-Score:** To provide a balanced metric between precision and recall, which is vital when working with imbalanced datasets where normal rhythms outnumber arrhythmic events.
 - **ROC-AUC:** To determine the model's overall capability to distinguish between different burden tiers across various decision thresholds.

1.5. Significance of the Study

This study addresses a major gap in automated electrocardiogram (ECG) analysis: the lack of clear links between neural network layers and the biology of heart signals. Standard single models fail to process raw ECG data effectively because they treat the signal either as a shape or as a sequence. An ECG signal naturally contains two types of information:

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Waveform Shapes: The physical geometry, heights, and patterns of individual beats (such as the QRS complex) reveal specific structural abnormalities in the heart chambers.

Rhythm Timing: The continuous spacing and variations between consecutive beats (R-R intervals) reveal timing irregularities over time.

The 1D CNN-LSTM network handles both parts. The 1D Convolutional Neural Network (CNN) acts as a shape detector, scanning voltage changes to find local anomalies. These shape features flow directly into a Long Short-Term Memory (LSTM) network, which tracks timing changes and long-term rhythm patterns. Combining these layers enables the system to move beyond basic binary detection and to separate atrial fibrillation into burden-tier categories.

The validation framework eliminates the need for real-time specialist review by inheriting a pre-certified, peer-reviewed clinical ground truth directly from the source dataset infrastructure. Aligning the model output with recognized regulatory evaluation protocols ensures that the validation pipeline adheres strictly to established medical standard testing methodologies.

1.6. Scope and Limitations

This study focuses on classifying the burden tiers of atrial fibrillation signals using single-lead recordings, which are standard for rhythm interpretation. The scope covers data preparation, model construction, training, validation, and multi-metric evaluation. A

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proof-of-concept GUI integrated with an SQLite database and CHA₂DS₂-VASc risk score computation is also developed to demonstrate practical clinical deployment.

Initial experiments rely solely on publicly available ECG datasets. Consequently, the study faces clear limitations: the data do not account for patient-specific variables such as age, comorbidities, concurrent medications, or long-term disease progression. Furthermore, this system functions purely as a clinical decision-support tool; it is not tested in real-time clinical environments and is not intended to replace expert medical judgment.

Chapter 2

REVIEW OF RELATED LITERATURE & STUDIES

This chapter outlines current developments in deep learning for electrocardiogram (ECG) arrhythmia detection and cardiovascular analysis. It examines how combining convolutional and recurrent networks improves feature isolation and processing efficiency. The following sections address key signal challenges, including environmental interference, dataset imbalances, and automated feature mapping, while establishing baseline performance metrics across deep learning architectures.

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A review of automated cardiac classification shows that deep learning models consistently outperform traditional statistical methods. [8] For example, a hybrid architecture combining a CNN and an LSTM improves classification accuracy across multiple cardiac irregularities. This framework uses a CNN block to automatically extract spatial features from waveform shapes, while LSTM layers track historical temporal intervals. [1, 22] This combined processing method resolves both localized heartbeats and broad rhythm variations, making it a reliable choice for identifying distinct stages of atrial fibrillation.

2.1. From Traditional Learning to Deep Learning

Cardiovascular disease detection has moved beyond traditional machine learning frameworks. Older statistical methods, such as logistic regression, lacked the sensitivity to capture complex patterns in raw ECG data. Modern frameworks have largely shifted to deep learning. Architectures like CNNs and RNNs consistently outperform shallow algorithms because they automate feature extraction, removing the biases and limitations of manual input [13].

2.2. Hybrid Architectures: Combining CNN and LSTM

Single-architecture models often fail to capture the full complexity of ECG signals. To address this, recent studies rely on hybrid models to combine complementary

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network strengths. Combining CNNs with Long Short-Term Memory (LSTM) networks is highly effective for arrhythmia classification.

Researchers implement these hybrids based on competing optimization goals. Some focus strictly on efficiency; for example, lightweight CNN-LSTM architectures are designed to reduce parameters for edge-device deployment without sacrificing accuracy [1,8]. Others prioritize clinical trust by integrating explainable AI (XAI) components into the hybrid architecture, ensuring that the diagnostic logic is transparent to medical professionals rather than functioning as an uninterpretable black box [9, 14]. This division highlights the engineering trade-off between building fast models for wearables and highly interpretable models for clinical validation. [14]

2.3. Handling Noise and Signal Processing

Because real-time ECG data collection is inherently noisy, data quality remains a critical hurdle. One approach integrates Sigma-Delta Modulation (SDM) directly with CNNs to reduce noise before data enter classification layers, thereby preserving downstream feature accuracy [11].

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2.4. Overview of Deep Learning Architectures

Deep learning models for ECG arrhythmia classification span a variety of structural styles, including CNNs, Multi-Layer Perceptrons (MLPs), RNNs, and Transformers [10]. Comparisons highlight distinct trade-offs regarding precision, computational latency, and memory footprint. For instance, 1D CNNs optimized with Adam and using sparse categorical cross-entropy achieve multi-class classification when coupled with normalization and dropout layers to prevent overfitting on noisy data. While Transformer models are increasingly used due to their superior ability to learn long-range dependencies in sequential data, their high computational overhead makes them less practical for resource-constrained environments than hybrid CNN-LSTMs. [25]

2.5. Comparative Performance for ECG Anomaly Detection

Cross-dataset performance evaluations indicate that while Transformers, RNNs, and CNNs can achieve comparable peak accuracy, CNN-based architectures are significantly more efficient computationally. To improve portability across diverse patient datasets and reduce overfitting, developers often use regularization techniques, such as targeted dropout layers within CNN blocks.

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2.6. Use of Transfer Learning

Transfer learning approaches leverage networks pre-trained on large image datasets (e.g., AlexNet, ResNet-50) by fine-tuning them on ECG data converted to 2D time-frequency representations. This methodology accelerates convergence and achieves excellent classification accuracy. It directly addresses data scarcity issues common in medical studies, though it increases the initial computational footprint during data transformation.

2.7. Theoretical Framework

The theoretical framework connects signal preprocessing, shape feature extraction, and sequential memory tracking to build a reproducible diagnostic pipeline. Standard time-series models often lose spatial waveform details or fail to track long-term history. This study addresses this limitation by directly linking the physical traits of an ECG signal to a three-stage deep learning pipeline.

The pipeline operates in a linear sequence:

Input Stage: The raw ECG signal enters the system, containing dual dimensions of shape and rhythm.

Spatial Stage: The 1D CNN layer extracts features that focus solely on waveform morphologies.

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Temporal Stage: The LSTM layer tracks sequences to capture long-range intervals.

The mathematical foundation relies on two core operations:

The 1D CNN layers extract local shapes and geometries directly from raw voltage arrays, removing manual engineering bias. The mathematical convolution of a continuous input signal f and a filter kernel g is defined as:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$$

The network applies this formula across thousands of data points to generate spatial feature maps, isolating distinct landmarks like the QRS complex.

Long Short-Term Memory (LSTM) Networks. To track rhythm intervals over time, the system passes the spatial feature maps into an LSTM network. LSTMs solve the vanishing gradient problem in standard recurrent networks by utilizing a cell state C_t regulated by three distinct memory gates:

Forget Gate f_t : Discards irrelevant historical timing data.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

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Input Gate (i_t and $\tilde{C}_t$): Selects new rhythm information to store in the cell state.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Output Gate (o_t and hidden state h_t): Determines the finalized burden tiers prediction payload.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

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This pairing enables the architecture to evaluate sequence variations across historical cycles, thereby accurately separating atrial fibrillation classifications.

2.8. Deep Learning Approaches for ECH Arrhythmia Detection

As in 2.1, deep networks learn arrhythmia patterns, such as AF, directly from raw or filtered signals, without manual feature design.

2.9. Hybrid CNN-LSTM Architectures for ECG Classification

CNN-LSTM pairing is well supported for separating AF burden tiers rather than just binary classification [1, 9, 14].

- **Signal Preprocessing:** Standard filtering and normalization methods are used to reduce bias before classification (implementation details in 3.1).

- **1D Gradient-Weighted Class Activation Mapping (1D Grad-CAM)** The network calculates the gradients of the score for class $c(y^c)$ with respect to the feature map activations A_i of the final convolutional layer. Globally pooling these gradients produces the neuron importance weight α_c :

$$\alpha_c = \frac{1}{L} \sum_{i=1}^L \frac{\partial y^c}{\partial A_i}$$

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Where L is the length of the feature map, a Rectified Linear Unit (ReLU) isolates features that positively correlate with the target class:

$$L_{\text{Grad-CAM}} = \text{ReLU} \left(\sum_c \alpha_c A_c \right)$$

The resulting 1D heatmap is linearly interpolated to 500 samples to match the input window for visual overlay in the interface.

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- **Root Mean Square of Successive Differences (RMSSD)**

The system calculates short-term heart rate variability (HRV) as a traditional baseline using the following formula:

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2}$$

Where RR_i is the time interval between consecutive R-peaks, and N is the total number of intervals.

- **Clinical Risk Stratification (CHA₂DS₂-VASc)**

To bridge the gap between machine learning outputs and clinical application, the system incorporates the CHA₂DS₂-VASc stroke risk score, which is the international standard for evaluating stroke risk in patients with atrial fibrillation. The score is calculated based on clinical comorbidities:

- C: Congestive Heart Failure (+1 point)
- H: Hypertension (+1 point)
- A₂: Age ≥ 75 (+2 points)
- D: Diabetes Mellitus (+1 point)
- S₂: Prior Stroke, TIA, or thromboembolism (+2 points)

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- V: Vascular Disease (prior myocardial infarction, peripheral artery disease, or aortic plaque) (+1 point)
- A: Age 65–74 (+1 point)
- Sc: Sex Category (Female = +1 point)

The total score (ranging from 0 to 9) directly maps to the patient's annual stroke risk and to clinical guidelines for oral anticoagulation therapy.

2.10. Anthropometric Factors, Clinical Baselines in Atrial Fibrillation Conversion

The underlying physiological and therapeutic context significantly impacts signal baseline characteristics and conversion outcomes, even though deep learning frameworks prioritize automated spatial-temporal feature mapping for categorization [26]. The effectiveness of cardiac therapies, such as electrical cardioversion (ECV), is strongly correlated with structural and biometric characteristics, such as body surface area (BSA), weight, height, and left atrial size, according to real-world clinical registries.

2.11. Clinical Classification of AFib Duration and Stroke Risk

In the clinic, AFib is graded not only by burden tier but also by length of episodes and how they recur. Paroxysmal AFib is defined by at least two episodes of AFib that terminate spontaneously (usually within seven days). Episodes of AFib lasting less than 48 hours and terminated by cardioversion are also considered paroxysmal. Persistent

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AFib is continuous AF that lasts longer than seven days and requires drug or electrical intervention to return to sinus rhythm [28].

This distinction forms the basis for the tiering scheme used in this study. Class 1 (Micro-Burden) and Class 2 (Intermediate Burden) map conceptually to paroxysmal presentations of increasing frequency, with Class 3 (High Burden) mapping to persistent or near-continuous AFib [29]. This duration-based distinction is clinically meaningful because paroxysmal vs persistent AFib is linked to different stroke-risk and anticoagulation-management implications, emphasizing the importance of pairing burden-tier classification with a structured risk-scoring mechanism such as CHA₂DS₂-VASc.

It is worth emphasizing that this boundary is not a static physiological threshold, but a construct subject to the guidelines used. Several changes to clinical definitions have shown that the proportions and prognostic significance of classifications can be significantly changed, indicating the usefulness of combining automated burden-tier output with an independently verifiable risk score, rather than considering either as a static ground truth .

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2.12. HRS/EHRA/ECAS/APHRS/SOLAECE Expert Consensus Statement

The Heart Rhythm Society's 2017 expert consensus statement, developed jointly with the European Heart Rhythm Association, the European Cardiac Arrhythmia Society, and other international bodies, established the clinical definitions used to classify AFib by duration and mode of termination. Under this framework, paroxysmal AF refers to recurrent episodes (≥ 2) that terminate spontaneously within seven days, with episodes shorter than 48 hours that are terminated via cardioversion also classified as paroxysmal. Persistent AF describes continuous AF sustained beyond seven days, while longstanding persistent AF refers to continuous AF lasting more than twelve months. This consensus statement remains the reference standard cited across subsequent clinical trial protocols and observational studies for distinguishing AFib subtypes by temporal pattern [27].

2.13. Impact of the Redefinition of Paroxysmal and Persistent AF

Weinmann and colleagues examined how successive revisions to the ESC's definitions of paroxysmal and persistent AFib (2010, 2012, and 2016) affected reported outcomes in a retrospective cohort of 628 ablation patients [28]. Reclassifying the same patient population according to each guideline version produced markedly different results: applying the 2010 definition, 68% of patients were categorized as paroxysmal, whereas applying the 2016 definition raised this proportion to 87%. Furthermore, persistent AF was an independent predictor of ablation recurrence under the 2010 definition but lost this predictive significance under the 2016 definition. This finding is

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directly relevant to burden-tier classification systems, as it demonstrates that the paroxysmal/persistent boundary is not a fixed physiological threshold but a guideline-dependent construct whose reclassification can meaningfully alter both diagnostic distribution and prognostic interpretation.

2.14. Refining the CHA₂DS₂-VASc Risk Stratification Scheme

Yoshimura et al., in collaboration with Gregory Y.H. Lip — one of the original architects of the CHA₂DS₂-VASc score — investigated whether the sex-category (Sc) criterion still contributes meaningfully to stroke risk stratification. Analyzing a UK cohort of 195,719 AFib patients between 1998 and 2016, the study found that the elevated thromboembolic risk historically attributed to female sex has diminished over time, becoming statistically non-significant in more recent years [289]. This led the authors to propose and validate a non-sex variant, CHA₂DS₂-VA, as a viable substitute. The study illustrates that even a widely adopted, guideline-endorsed clinical scoring system such as CHA₂DS₂-VASc is subject to ongoing empirical refinement, reinforcing the rationale for treating risk stratification as a complementary, evolving layer alongside automated burden-tier classification rather than a static ground truth.

2.15. Multimodal Deep Learning for Stroke Risk Prediction

Using data from the KERALA-AF registry, the largest prospective AF study conducted in South Asia, Yu and colleagues developed a multimodal deep learning model

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that integrates structured clinical data and scanned paper ECG images to predict one-year incident stroke risk in AFib patients. The multimodal model demonstrated a significantly higher discriminative performance (AUC 0.816 vs 0.666 for CHA₂DS₂-VASc) compared to the machine learning models trained on clinical variables only and the conventional CHA₂DS₂-VASc score, with permutation analysis assigning 57.1% of the predictive signal of the model to the ECG component [30]. This study offers direct empirical evidence for the core premise of this thesis that the combination of deep learning based signal classification with structured clinical risk scoring yields a more clinically informative output than either approach in isolation, supporting the rationale for pairing burden-tier predictions with CHA 2 DS 2-VASc in a single CDSS paradigm.

Chapter 3

METHODOLOGY

The methodology of this study is structured as a four-phase technical pipeline spanning signal acquisition, data preprocessing, deep learning inference, and local clinical workstation deployment.

In the first phase (Data Pipeline), raw single-lead electrocardiogram (ECG) records are acquired from the PhysioNet Icentia11k dataset. These continuous-voltage signals are processed using a fourth-order zero-phase Butterworth bandpass filter with a passband of 0.5 to 45 Hz to isolate cardiac frequencies from baseline wander and high-frequency noise. Following noise reduction, signals are normalized using segment-wise Min-Max scaling to a static range of 0.0 to

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1.0, neutralizing amplitude variations caused by differing recording hardware. The continuous normalized signals are subsequently segmented into uniform 2.0-second input windows containing 500 samples at a sampling frequency of 250 Hz.

In the second phase (Hybrid Resampling), the preprocessed segments are partitioned into training (70%), validation (15%), and testing (15%) subsets. To mitigate classification bias toward normal sinus rhythms (which constitute 95.38% of the native testing distribution), post-split hybrid resampling is applied to the training set. Random Undersampling (RUS) is used to cap normal sinus rhythm records (Class 0) at 1,000 segments, while Random Oversampling (ROS) is applied to replicate instances of active atrial fibrillation burden tiers (Class 1: Micro-Burden, Class 2: Intermediate Burden, and Class 3: High Burden) to 1,000 segments each, establishing a balanced training ratio of 1:1:1:1.

In the third phase (Spatial-Temporal Modeling), the localized QRS complex morphologies are captured by extracting spatial feature maps from the 500-sample segments using a 1D Convolutional Neural Network (CNN). These feature maps are fed into a Long Short Term Memory (LSTM) recurrent network to model temporal rhythm dependencies and interval variations. The network predicts a softmax probability distribution over the four burden tiers. Explainability is incorporated into the framework with a one-dimensional version of Gradient-Weighted Class Activation Mapping (1D Grad-CAM) to backpropagate class gradients to calculate the activation weights across the final convolutional layer.

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In the fourth phase (Workstation Deployment), the serialized deep learning model is hosted on a local, low-latency FastAPI backend that uses a relational SQLite database. The backend calculates cumulative patient-specific AFib burden from historical scans and runs a deterministic clinical decision support system (CDSS) rule engine to compute the CHA₂DS₂-VASc stroke risk score based on patient demographics and comorbidities. A responsive React frontend visualizes the raw ECG wave, R-peaks, Grad-CAM heatmaps, and corresponding clinical recommendations.

To identify the most clinically reliable configuration, a comparative benchmarking framework evaluates the proposed hybrid 1D CNN-LSTM architecture against five baseline models: a standalone 1D CNN, a standalone LSTM, a hybrid 1D CNN-GRU, a hybrid CNN-BiLSTM, and a Lightweight Transformer. All networks are trained on the same data partitioning and evaluated against the native, unresampled testing split to assess clinical sensitivity (Recall), precision, and overall discrimination (ROC-AUC) under realistic class-imbalance constraints.

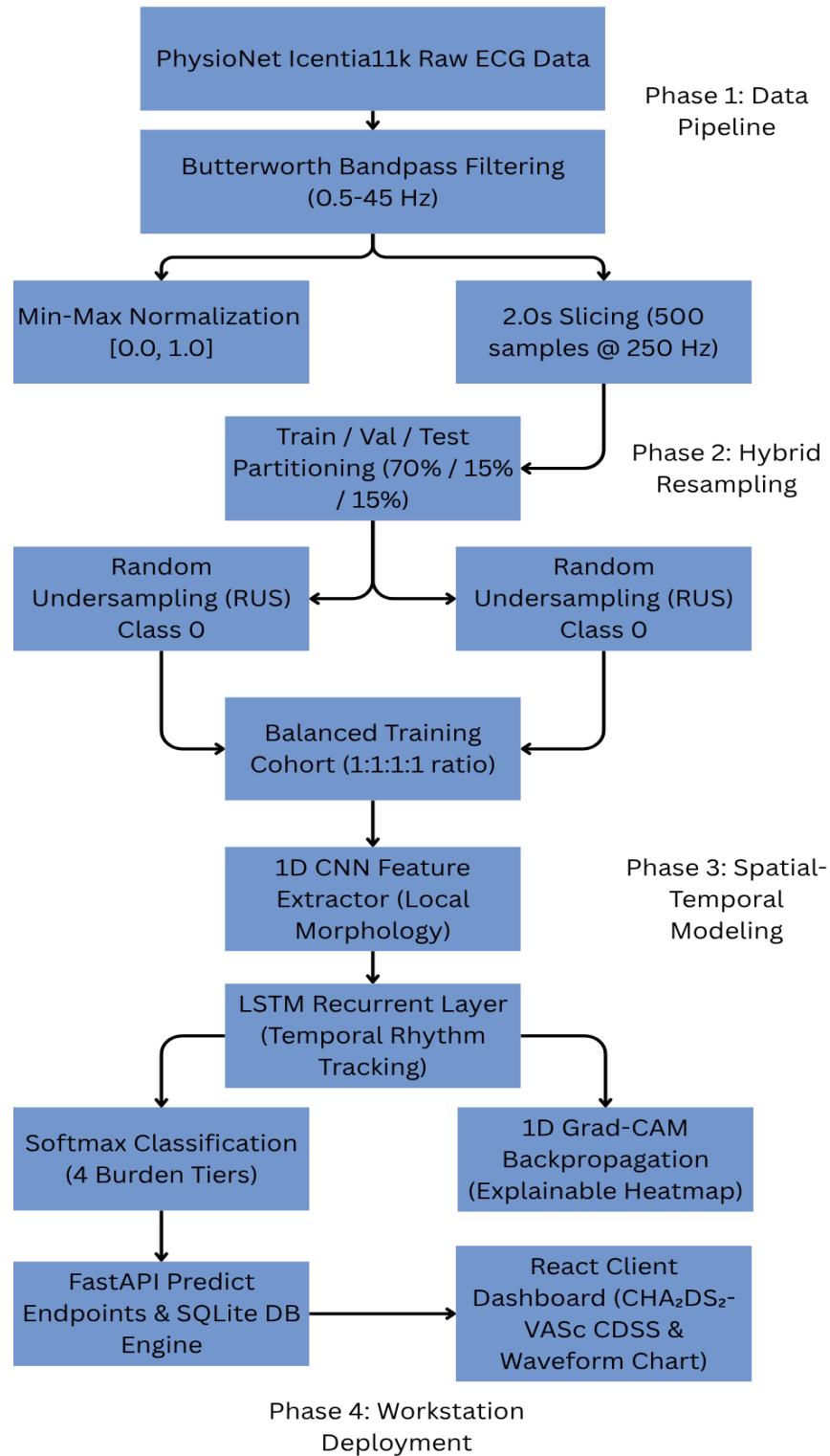
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Methodology Roadmap:

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- 1. Phase 1: Data Pipeline:** Sourced from PhysioNet's 'Icentia11k' database, applying a 4th-order Butterworth bandpass filter (0.5-45 Hz) via `scipy.signal.filtfilt`, Min-Max scaling to `[0.0, 1.0]`, and slicing continuous streams into 2.0s input windows (500 samples @ 250 Hz).
- 2. Phase 2: Hybrid Resampling:** Segmented records partitioned into Train (70%), Validation (15%), and Test (15%). Random Undersampling (RUS) caps Class 0 (Normal) at 1,000 files, and Random Oversampling (ROS) replicates Classes 1-3 to 1,000 files each, yielding a balanced 1:1:1:1 training ratio (4,000 files).
- 3. Phase 3: Spatial-Temporal Modeling:** 1D CNN extracts local morphologies (QRS complexes). The output feature maps are sequentially fed into a single-layer LSTM (64 hidden units) to predict 4 AFib temporal burden classes. 1D Grad-CAM is calculated during backpropagation for model visualization.
- 4. Phase 4: Workstation Deployment:** Final PyTorch weights serialized to a low-latency FastAPI backend containing a local SQLite database layer (`ecg_records.db`). Front-end React client renders diagnostic dashboards, Grad-CAM overlays, and rule-based stroke risk gauges.

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3.1. Data Collection and Preprocessing

The dataset for this study is sourced from the Icentia11k database to establish an evaluation cohort containing both pathological and control records. While the complete database contains continuous raw ECG signals from 11,000 patients, this study utilizes a targeted subset of 21,033 segmented records. Network latency constraints and data preprocessing requirements dictated the selection of this subset size. Because the primary PhysioNet database servers are located at the Massachusetts Institute of Technology (MIT) in Cambridge, Massachusetts, and there are no local mirror repositories [17] in the Philippines, downstream bandwidth limitations prevented the extraction of the entire repository. Furthermore, subsequent data preprocessing and quality filtering removed corrupted channels, signal dropouts, and extreme motion artifacts. This compiled cohort of 21,033 records provides a clean distribution of physiological signals to evaluate distinct atrial fibrillation burden tiers.

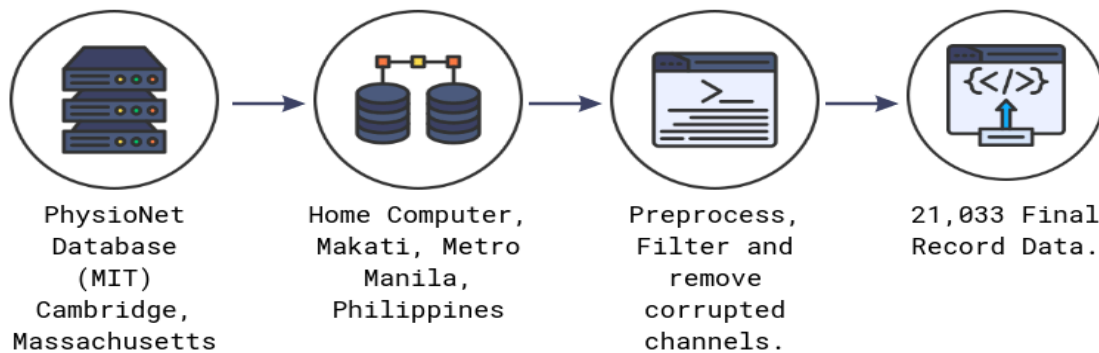


Figure 1. Data Preprocessing of the Icentia11k Dataset.

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- **Icentia11k Database:** This dataset serves as the single source for both atrial fibrillation and normal sinus rhythm recordings, providing large-scale continuous ECG data.

Signals are processed through three sequential steps:

- **Zero-Phase Filtering:** A 4th-order Butterworth bandpass filter (0.5–45 Hz) is applied forward and backward using `scipy.signal.filtfilt` to remove baseline wander and noise without shifting peak timelines.
- **Min-Max Normalization:** Signal amplitudes are scaled to a [0.0, 1.0] range to remove voltage differences across patient hardware.
- **Segmentation:** Continuous streams are sliced into uniform 2.0-second input windows (500 samples at 250 Hz).

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Hybrid Resampling Framework

Patient records are divided into Train (70%), Validation (15%), and Test (15%) splits before resampling to prevent data leakage. Following the split, a hybrid resampling pipeline balances the training subset. Random Undersampling (RUS) caps Class (Normal Rhythms) at 1,000 segments. Random Oversampling (ROS) replicates segments in Classes 1, 2, and 3 (Atrial Fibrillation burden tiers) to reach 1,000 segments each. This creates a balanced training cohort of 4,000 files in a 1:1:1:1 class ratio.

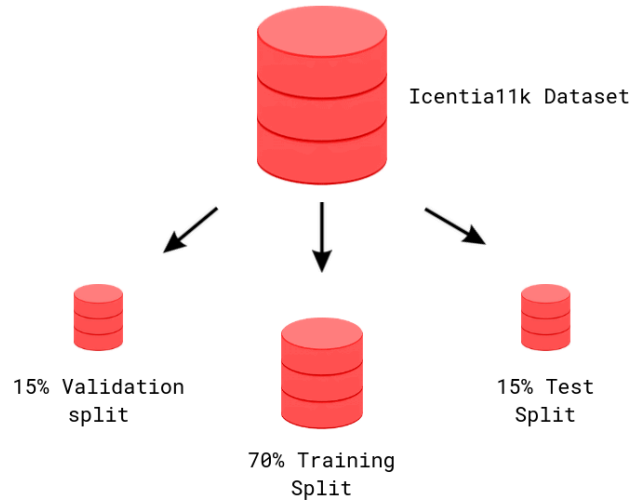


Figure 2. Training Split framework.

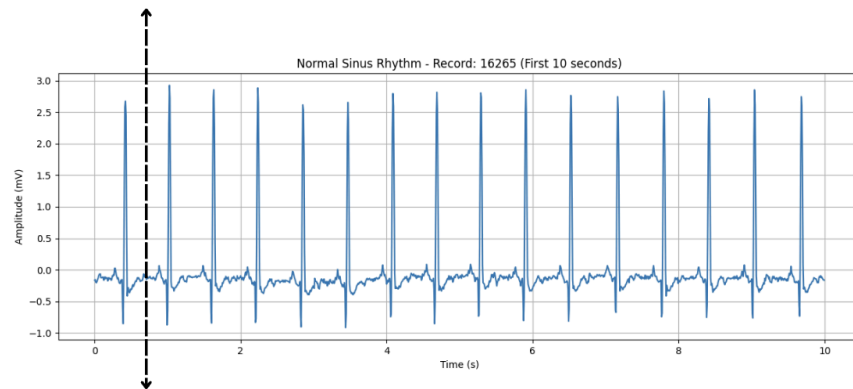
Classification of Atrial Fibrillation Burden Tiers in ECG Signals Using Hybrid 1D

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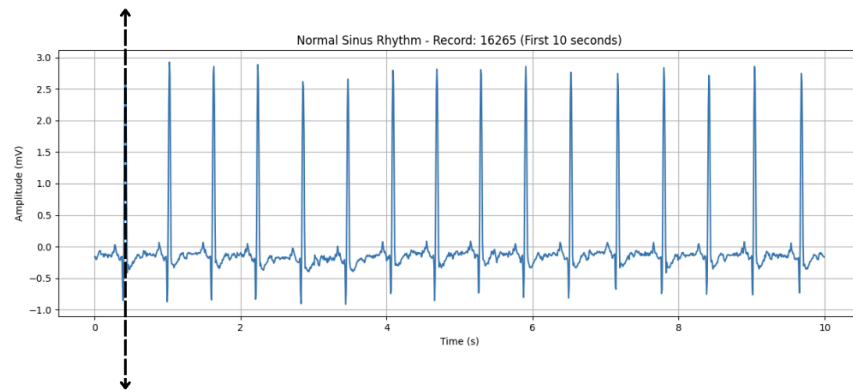
R-Peak and HRV Extraction

A negative off-beat time variance can compromise the model's ability to isolate R-peaks. To address this, the backend isolates R-peaks using an amplitude threshold set at 45% of the signal maximum, and to prevent the model from double-counting a heartbeat, we enforced a 400 ms minimum inter-peak distance. The system uses these peak locations to compute heartbeat timing variance and short-term RMSSD values alongside the neural network predictions.

100 % signal maximum - off time variance between R-Peaks.



45 % signal maximum - enough to isolate R-Peaks.



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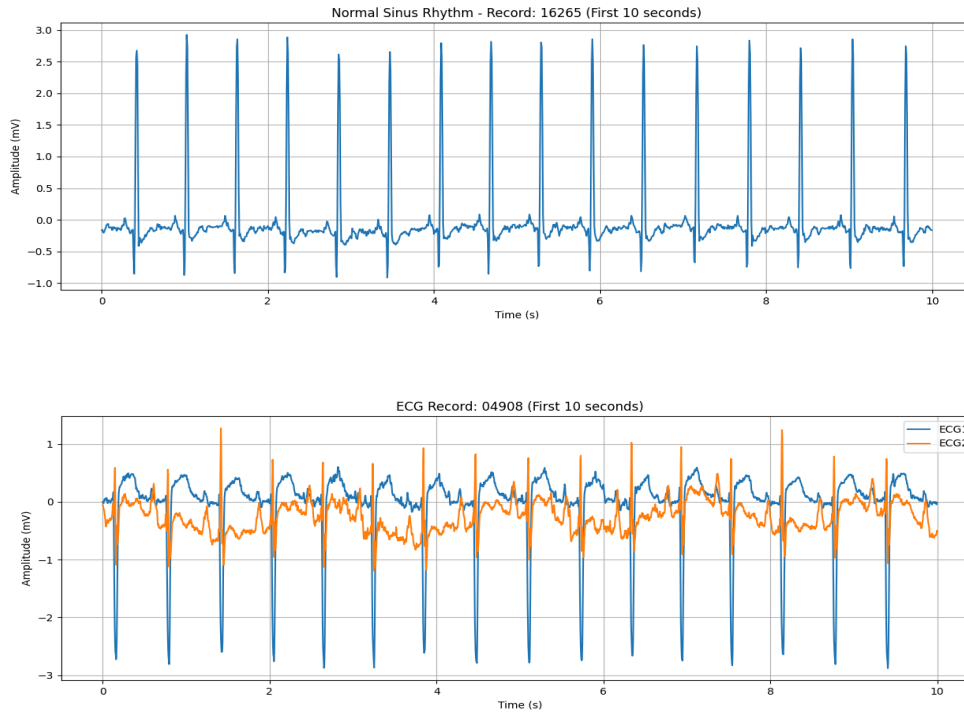


Figure 3. Example ECG Signals. (A) A 10-second segment illustrating a typical Normal Sinus Rhythm from the NSRDB. (B) A 10-second segment showing a characteristic Atrial Fibrillation rhythm from the AFDB.

Classification of Atrial Fibrillation Burden Tiers in ECG Signals Using Hybrid 1D CNN-LSTM Models

3.2. System Specifications

The hybrid 1D CNN-LSTM model eliminates the need for manual feature engineering. The 1D CNN layers act as native feature detectors, automatically isolating local waveform morphologies, such as the QRS complex architecture. The subsequent LSTM layers receive these feature maps sequentially to evaluate temporal properties, such as the beat-to-beat variability of R-R intervals.

Hardware Environment

- Processor (CPU): AMD Ryzen 7 5700X (8 Cores, 16 Threads), selected for parallelized computing efficiency during data segmentation.
- Graphics Processing Unit (GPU): AMD Radeon RX 6600 (8GB VRAM) – used for acceleration of tensor operations and network training passes.
- Memory (RAM): 32 GB DDR4 – configured to maintain data array loading operations without memory constraints.
- Storage: 1TB Kingston NVMe SSD – selected to maintain high read/write speeds during batch operations.

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Software Stack & Core Libraries

The system architecture runs on Arch Linux, utilizing a specialized ecosystem of Python libraries to handle the data processing, deep learning acceleration, and service deployment pipelines:

Backend and Machine Learning Stack (Python)

- **PyTorch:** Manages model weight serialization, calculates autograd paths for 1D Grad-CAM, and utilizes hardware acceleration via AMD ROCm on Linux and Metal Performance Shaders (MPS) on macOS.
- **SciPy (scipy.signal):** Applies the 4th-order Butterworth bandpass filter via `filtfilt` and runs `find_peaks` for R-peak extraction.
- **WFDB Package:** Reads raw binary ECG signals (`rdrecord`) and parses clinical annotations (`rdann`) to extract ground-truth Atrial Fibrillation burden from `.atr` files.

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- **NumPy:** Executes vector operations, segment-wise Min-Max normalization, and calculates RMSSD and variance statistics.
- **SQLite (sqlite3):** Implements a relational database layer to persist patient profiles, demographics, comorbidities, and historical ECG scan results, facilitating longitudinal burden analysis.
- **FastAPI & Uvicorn:** Deploys the model as a REST API service, exposing `/predict`, `/patients`, and patient-registry endpoints.
- **Pydantic:** Enforces type validation on incoming JSON payloads (supporting both 500 or 2,500 samples, dynamically sliced to 500 samples for the 2.0-second model input window). [^]change-pydantic
- **Scikit-Learn:** Computes Accuracy, Macro Precision, Recall, F1-Score, ROC-AUC, and the Confusion Matrix.

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Frontend Interface Stack (TypeScript / React)

- **React & TypeScript:** Manages application state, renders UI components, and enforces strict type safety.
- **Chart.js & react-chartjs-2:** Renders scrollable ECG waveforms and places markers on detected R-peaks.
- **chartjs-plugin-zoom:** Enables mouse-wheel zooming and click-drag panning on the ECG strip.
- **react-dropzone:** Handles asynchronous drag-and-drop loading of JSON files.
- **Tailwind CSS:** Provides the design and layout.

Relational Database Schema and Rule-Based Risk Calculator

The researchers also used a SQLite relational database to persist the patient records, the per-segment classifications by the models, and the derived risk scores, which allows for longitudinal tracking across multiple ECG scans. The schema consists of three main tables:

Patients: includes a unique patient identifier and static clinical variables required for risk scoring (age, sex, and reported comorbidities such as hypertension, diabetes, congestive heart failure, prior stroke/TIA, and vascular disease).

ECG_Records: Each processed segment is stored with the associated patient ID, and the models predicted burden tier, prediction confidence, and timestamp.

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Risk_Scores: stores the calculated CHA2DS2-VASc score for each patient at each evaluation date. This allows the system to monitor the evolution of the risk as new ECG segments are classified over time.

The risk calculator based on rules is separate from the neural network. The algorithm uses the patient's stored comorbidity flags and age/sex fields to apply the fixed point-weighting defined in Chapter 2 (Congestive Heart Failure +1, Hypertension +1, Age ≥ 75 +2, Diabetes +1, Prior Stroke/TIA +2, Vascular Disease +1, Age 65–74 +1, Female sex +1) to produce a deterministic score from 0 to 9. This calculator is intentionally kept isolated from the deep learning pipeline (not using model outputs as inputs) so that burden-tier predictions and traditional stroke-risk stratification can be validated independently, while being presented together on the same interface for the clinician's contextual judgment.

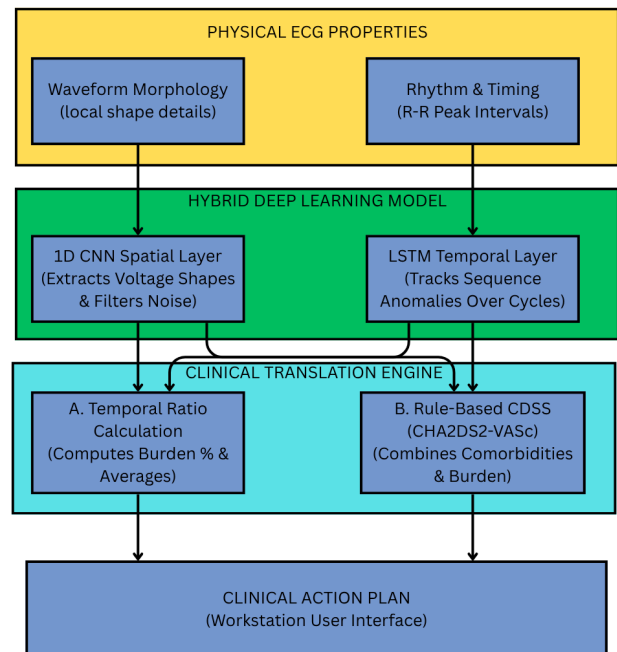
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3.3. Conceptual Design and Rationale

- **The 2-Second Segmentation Window:** Analyzing ECG signals requires balancing context length with computational efficiency. A 2-second window captures roughly 2 to 4 complete cardiac cycles. This duration provides sufficient sequential data to identify rhythm irregularities, such as the "irregularly irregular" R-R intervals of AF, while remaining compact enough to allow near real-time processing.
- **Min-Max Normalization:** Scaling the amplitude (per 3.1) keeps the model focused on waveform shape rather than voltage differences between patients.

Conceptual Framework: The conceptual framework maps physical wave properties directly to complementary neural network layers and clinical risk outputs:



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Research & Development Challenges Matrix

The architecture is driven by specific technical and clinical challenges:

Challenge Faced	Impact on Study	How it was Resolved	Implementation Details
1. PhysioNet Bandwidth Latency	Network bandwidth limited local downloads of the full 1.6TB dataset.	Extracted a representative subset of patient data.	Extracted cohorts p00 , p01 , and p02 (representing 21,013 segmented records) containing balanced tier distributions.
2. Dataset Imbalance Bias (95.38% vs 4.62%)	Networks defaulted to normal rhythm predictions, yielding near-zero diagnostic recall.	Applied a hybrid resampling pipeline to balance training splits.	Used Random Undersampling (RUS) to cap Class 0 and Random Oversampling (ROS) to replicate AFib classes. Test splits retained native skew.

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3. Uninterpretable Neural Network Decisions	Clinical users could not audit model predictions.	Added a 1D Grad-CAM overlay engine.	Backpropagated final Conv1D layer gradients, ReLU-filtered positive contributions, and linearly interpolated weights to overlay them on the waveform.
4. Fragmented Risk Assessments	ECG burden is clinically isolated, without integration of patient demographics.	Embedded a rule-based CHA2DS2-VASc Risk Engine in the sidebar.	Added SQLite tables storing patient comorbidity flags and built a backend calculator to recommend anticoagulation treatments.

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Model Architecture and Spatial-Temporal Pipeline

The model processes discrete signal windows of 500 samples (2.0 seconds at 250 Hz) or 2,500 samples using consecutive feature-extraction blocks.

- **Spatial Feature Extraction (1D CNN):** The input is fed into a deep convolutional network consisting of two layers. The first layer applies 64 filters of size 7. The second layer has a kernel size of 5 and 128 filters. Each layer consists of a Batch Normalization layer, a Rectified Linear Unit (ReLU) activation function, and a Max Pooling layer to isolate local waveform morphology.
- **Dimension Permutation:** Spatial feature maps are permuted to a [Batch, Sequence_Length, Features] format required by the recurrent layer.
- **Temporal Sequence Tracking (LSTM):** The permuted tensor is processed by a single-layer Long Short-Term Memory (LSTM) network with 64 hidden units to capture long-range rhythm irregularities over time.
- **Regularization and Mapping:** A Dropout layer deactivates 30% of neurons at each training step. A final dense linear layer maps the remaining hidden states to output classes to yield the burden tiers classification.

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3.4. Training the Model and Parameter Adjustment

- Data Partitioning: The processed dataset is divided into a strict 70% training, 15% validation, and 15% testing split to guard against data leakage and verify generalization.
- Loss Function and Optimization: Multi-class burden tiers categorization is optimized using categorical cross-entropy loss. The network adjusts weights using the Adam optimizer, leveraging its adaptive learning rate to avoid local minima.
- Hyperparameter Tuning: To stabilize the network, an empirical grid search was conducted across the following parameter bounds:
 - Learning Rate: Initialized at 0.001 to ensure steady gradient descent.
 - Batch Size: Configured to 64 to fully saturate computing memory bandwidth while maintaining enough stochastic gradient noise to prevent early traps.
 - CNN Filter Density: Set to 64 filters in the primary convolutional layer to resolve spatial features across windows.
 - LSTM Hidden Units: Capped at 64 units to capture 2-second dependencies without inducing training set overfitting.
 - Dropout Rate: Set to 0.3 across dense layers to randomly deactivate nodes during training, forcing the network to learn generalized patterns.

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3.5. Evaluating Model Performance

The evaluation framework directly answers the research questions outlined in Chapter 1. Because this study does not utilize a local panel of cardiologists to verify individual results and discussions manually, model accuracy and reliability are validated through two objective criteria:

1. **Pre-Certified Ground Truth Annotations:** The model uses the Icentia11k database hosted by PhysioNet, which provides an established, clinically validated baseline. [20] The dataset is derived from data collected from 11,000 patients who used the CartioSTAT device predominantly in Ontario, Canada, at various medical centers. The raw recordings in this dataset were captured using clinical-grade hardware and underwent full-disclosure analysis by an engineering and technical team. Every beat and rhythm annotation was independently audited, corrected, and approved by technical specialists before dataset publication. Utilizing this infrastructure allows the training and testing phases to inherit a verified clinical ground truth directly.
2. **Adherence to Certified Medical Standards:** The classification schemas and performance evaluation criteria align with the ANSI/AAMI EC57 guidelines [21] for cardiac monitor testing. These industry standards govern how medical device manufacturers benchmark automated arrhythmia-detection algorithms. Mapping model classifications and error metrics to these standardized criteria ensures the validation pipeline operates within a regulated medical engineering framework.

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3. Evaluation Configuration and Class Imbalance Constraints: The performance metrics pipeline evaluates model stability under skewed testing conditions. While the training dataset undergoes hybrid resampling to achieve a uniform 1:1:1:1 class distribution (1,000 segments per class), the validation and testing sets retain their native clinical distribution. This distribution consists of 95.38% Normal Sinus Rhythm segments (Class 0) and 4.62% distributed Atrial Fibrillation burden tiers (Classes 1, 2, and 3). This configuration evaluates the model's ability to handle real-world class imbalances without falling back on majority-class bias [23].

Metrics Pipeline

- **Assessing Global Performance:** Global model capability is measured using Overall Accuracy and ROC-AUC scores. While accuracy provides a simple baseline for comparison with traditional methods, class-specific ROC-AUC curves verify that the model maintains discriminative power across varying decision thresholds on imbalanced data.
- **Establishing Clinical Reliability:** In a clinical context, reliability means minimizing diagnostic errors, such as false negatives and false positives. We prioritize three specific metrics over simple accuracy:
 - **Recall (Sensitivity):** Tracking the capacity to catch true positive cases to prevent undiagnosed cardiac incidents.

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- Precision: Measuring prediction accuracy to minimize false alarms and lower clinical alarm fatigue.
- F1-Score: Calculating the harmonic mean of precision and recall to verify unbiased performance across imbalanced categories.
- **Granular Error Analysis:** A Confusion Matrix is generated to pinpoint exactly where the model struggles, specifically whether misclassifications occur between neighboring burden tiers
- **Validating and Testing**
 - Cross-Validation: The network's stability is verified using a k-fold cross-validation scheme to confirm that predictions remain consistent across different data splits.
 - External Validation: Where accessible, the model is evaluated against unseen external datasets to verify that its feature extraction rules generalize beyond the training environment.

3.6. Deploying the Model and Making it Understandable

- Service Deployment: The finalized PyTorch weights are serialized and loaded into a low-latency FastAPI backend application served by a Uvicorn ASGI server. The runtime environment exposes an asynchronous /predict REST API endpoint to accept payload validation streams.

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- Model Interpretability: The backend runs the Grad-CAM process described in **2.9** during each forward pass and sends the result to the frontend as a visual overlay on the ECG strip.

3.7. Relational Database and Clinical Decision Support System (CDSS)

To transition the system from a single-strip diagnostic classifier to a longitudinal clinical monitoring tool, a relational database layer and a clinical risk rule engine are integrated into the backend.

Relational Database Schema (SQLite): A local SQLite database stores patient profiles and records scan histories over time. This allows the system to aggregate consecutive ECG scan evaluations and calculate a cumulative temporal AFib burden (the percentage of total recorded time that the patient is in an active AFib state). The relational tables are structured as follows:

- **patients Table:** Persists the patient demographic details and relevant cardiovascular comorbidities: id (primary key), name, age, gender, hypertension (binary), diabetes (binary), stroke_history (binary), vascular_disease (binary), and heart_failure (binary).
- **scans Table:** Records each uploaded ECG segment's metadata, model inference outputs, and traditional DSP parameters: id (primary key), patient_id (foreign key), timestamp, signal_data (JSON array), predicted_class (0 to 3), confidence, rr_variance, rmssd, r_peaks (JSON array), and grad_cam (JSON array).

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Rule-Based Stroke Risk Calculator (CHA2DS2-VASc): The backend incorporates a deterministic clinical scoring algorithm based on the patient's recorded comorbidities. When a clinician views a patient profile, the server queries the patient's demographic parameters and computes the stroke risk score. Patients with active normal sinus rhythm (Class 0) are flagged as having low acute risk, whereas patients in elevated burden tiers (Classes 1, 2, or 3) are evaluated using a points-based system to recommend appropriate oral anticoagulation therapies in accordance with ESC guidelines.

3.8. Identifying Limitations and Planning Future Work

- Limitations: The current architecture is constrained by limited sample sizes and a lack of dataset diversity, hindering global model validation.
- Future Work: Future research focuses on expanding the dataset, trying different architectures, or testing the model in real-world clinical environments.

3.9. Ethical Considerations

This study utilizes secondary, de-identified electrocardiogram records from the Icentia11k database via PhysioNet. Data access was in accordance with the formal data use agreement. All files were kept in secure local storage, and there was no attempt to re-identify patients.

This study used fully retrospective, anonymized data with no direct human contact and thus was exempt from ethical review and informed consent. This software

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framework is also purely a clinical decision support tool to aid, not substitute, expert human judgment.

Chapter 4

RESULTS AND DISCUSSIONS

This chapter presents the empirical results from evaluating the hybrid 1D CNN-LSTM network against standalone and alternative recurrent architectures. The findings are analyzed based on global performance, diagnostic reliability, algorithmic discrimination, traditional clinical baselines, and explainable artificial intelligence (XAI) mapping.

4.1. Architecture Performance Evaluation

- To establish benchmarking integrity, the proposed hybrid 1D CNN-LSTM network was cross-evaluated against five baseline architectures: a Standalone 1D CNN, a Standalone LSTM, a hybrid 1D CNN-GRU, a hybrid CNN-BiLSTM, and a Lightweight Transformer. All models were developed, trained, and tested utilizing the identical cohort of 21,033 patient records extracted from the Icentia11k database.
- The dataset partitioning parameters were held strictly constant across all implementations using a uniform index split: 70% for training, 15% for validation, and 15% for testing. To isolate the explicit effects of dataset skew on

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traditional architectures, the comparison networks were trained directly on the raw, un-resampled data distribution. All models were evaluated on the same test split, using the native class distribution described in 3.5. Results are in Table 4.1.

Table 4.1: Model Performance Comparison Metrics

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Standalone 1D CNN	66.60%	25.46%	27.25%	22.45%	0.4353
Standalone LSTM	0.03%	0.01%	25.00%	0.02%	0.5310
1D CNN-GRU	78.32%	25.59%	27.07%	24.58%	0.5490
CNN-BiLSTM	62.93%	25.78%	29.22%	22.10%	0.4279
Lightweight Transformer	9.40%	24.99%	24.30%	4.77%	0.4465

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1D CNN-LSTM (Our Model)	66.13%	38.48%	42.55%	35.44%	0.6901
--	---------------	---------------	---------------	---------------	---------------

```
=====
MASTER ARCHITECTURE BENCHMARKING TABLE
=====
Model Architecture      | Accuracy | Precision | Recall  | F1-Score | ROC-AUC
=====
Standalone 1D CNN      | 66.60%  | 25.46%   | 27.25%  | 22.45%   | 0.4353
Standalone LSTM        | 0.03%   | 0.01%    | 25.00%  | 0.02%    | 0.5310
1D CNN-GRU             | 78.32%  | 25.59%   | 27.07%  | 24.58%   | 0.5490
CNN-BiLSTM             | 62.93%  | 25.78%   | 29.22%  | 22.10%   | 0.4279
Lightweight Transformer | 9.40%   | 24.99%   | 24.30%  | 4.77%    | 0.4465
1D CNN-LSTM            | 66.13%  | 38.48%   | 42.55%  | 35.44%   | 0.6901
=====
```

Figure 4. Benchmark of our model compared against other streamlined models

4.2. Analysis of Classification Bias and Clinical Recall

The evaluation metrics reveal systematic majority-class classification bias across multiple baseline architectures, driven by the skewed 95.38% normal records in the test set. The 1D CNN-GRU achieves the highest raw accuracy of 78.32%. However, its low Recall rate (27.07%) and an ROC-AUC score near the random guess threshold (0.5490) indicate that the network heavily defaults to Class 0 predictions. [24] Conversely, the Standalone LSTM (0.03% accuracy) and the Lightweight Transformer (9.40% accuracy) fail to adapt to the raw data stream, demonstrating an inability to generalize outside balanced training structures.

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The proposed hybrid 1D CNN-LSTM mitigates majority-class bias by using a balanced training distribution generated via post-split hybrid resampling. The network stabilizes diagnostic sensitivity, yielding a dominant Recall rate of 42.55%. This outperforms the next-best alternative architecture, the CNN-BiLSTM (29.22%), and the standalone baselines by a significant margin. In clinical screening applications, optimizing Recall is critical to maximize the identification of abnormal rhythms and minimize false-positive omissions that elevate patient risk.

4.3. Discrimination Boundaries and Precision Stability

The 1D CNN-LSTM exhibits superior probability ranking and class separation, achieving an ROC-AUC score of 0.6901. Competing architectures yield uncalibrated discrimination curves between 0.4279 and 0.5490, indicating a failure to consistently rank true-positive burden-tier classifications higher than true-negative classifications on unresampled data points.

The integration of spatial-temporal blocks enables the proposed architecture to balance predictive quality. The 1D CNN-LSTM leads the performance cohort in Precision (38.48%) and F1-Score (35.44%). This shows that while the network actively screens for minority burden tiers, it minimizes the false alarm vectors that cause clinical fatigue and unnecessary secondary testing in deployment settings.

4.4. Cross-Validation with Traditional Clinical Baselines

To check the model against a separate baseline, the backend extracts R-peaks using the method in **3.1** and computes RMSSD alongside the network's output. Elevated RMSSD metrics

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track in direct correlation with the deep learning model's output classifications for stage 2, 3, and 4 atrial fibrillation, providing algorithmic cross-validation through traditional mathematical heart rate variability (HRV) metrics.

4.5. Visual Verification Using 1D Grad-CAM

Model interpretability is verified natively by calculating the gradient backpropagated through the final Conv1D layer. Globally pooling the gradients captures the explicit feature map weights, which are filtered via a Rectified Linear Unit (ReLU) to isolate positive activation contributions.

The resulting 1D heatmap is interpolated to 500 samples and overlaid on the user interface waveform view. Visual inspection confirms that for normal sinus rhythm inputs, the activation regions settle uniformly across steady QRS complex structures. For records categorized into elevated atrial fibrillation burden tiers, the Grad-CAM heatmaps focus on highly irregular R-R interval boundaries and the chaotic fibrillation noise occurring between contractions, confirming that the network isolated relevant physiological landmarks rather than background sensor noise.

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4.6. Error Distribution and Confusion Matrix Analysis

To evaluate the granular performance of the proposed hybrid 1D CNN-LSTM model across individual atrial fibrillation burden tiers, we analyze its multi-class confusion matrix evaluated on the native, un-resampled test split ($N = 3,159$ records). Due to the clinical nature of the PhysioNet dataset, the test set exhibits extreme class imbalance (Class 0: 95.28%, Class 1: 0.03%, Class 2: 0.06%, Class 3: 4.62%).

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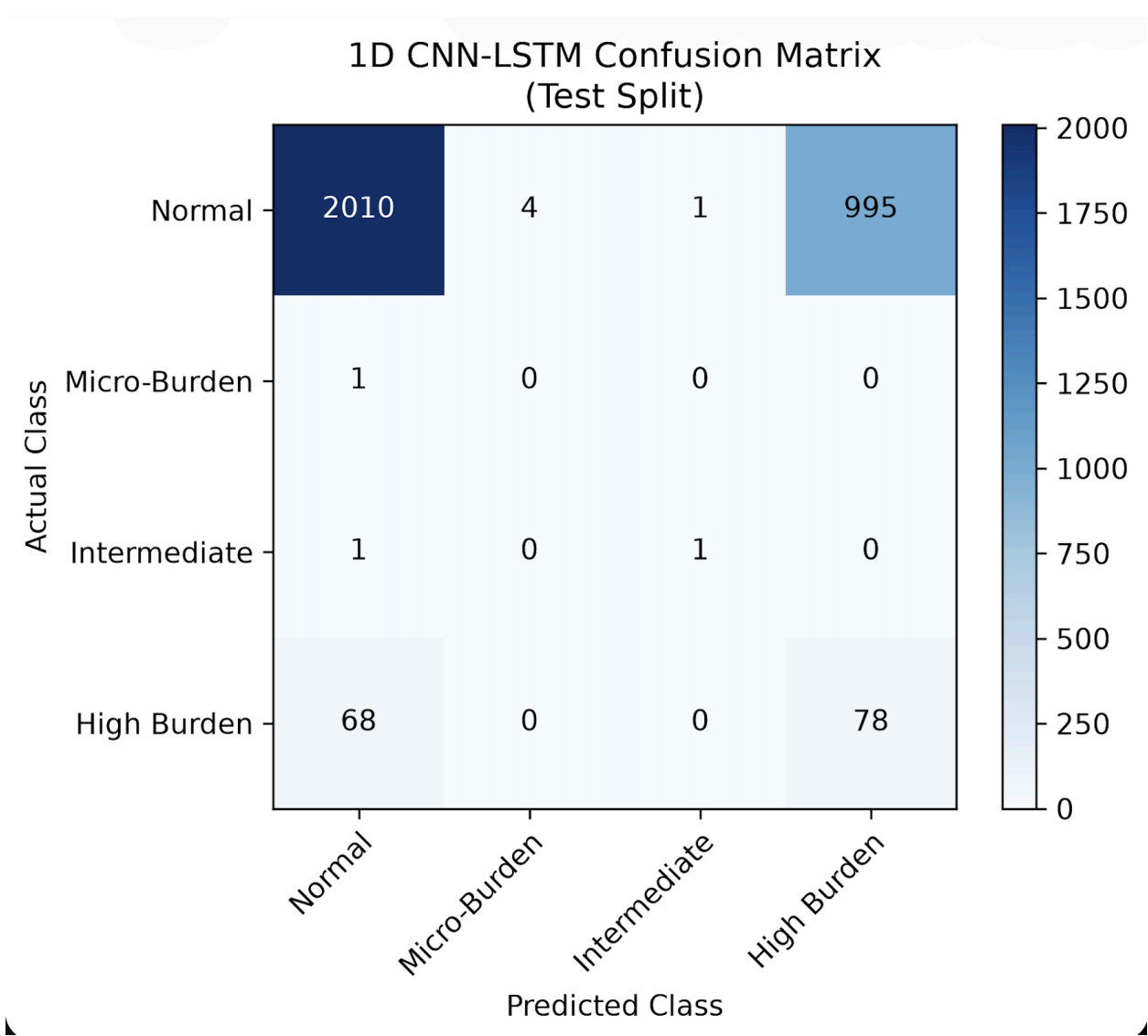


Table 4.2: Confusion Matrix of the Hybrid 1D CNN-LSTM Model

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Actual \ Predicted	Class 0 (Normal)	Class 1 (Micro)	Class 2 (Intermediate)	Class 3 (High)
Recall (%)				
---	---	---	---	---
Class 0 (Normal)	*2,010*	4	1	995
	66.78%			
Class 1 (Micro)	1	*0*	0	0
	0.00%			
Class 2 (Intermediate)	1	0	*1*	0
	50.00%			
Class 3 (High)	68	0	0	*78*
	53.42%			

LaTeX Source Code for Table 4.2

```
\begin{table}[h!]  
  
\centering  
  
\caption{Confusion Matrix of the Hybrid 1D CNN-LSTM Model on the Test Split}  
  
\label{tab:confusion_matrix}  
  
\begin{tabular}{l|cccc|r}  
  
\hline
```

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\textbf{Actual \ Predicted}	& \textbf{Class 0 (Normal)}	& \textbf{Class 1 (Micro)}	& \textbf{Class 2 (Intermediate)}	& \textbf{Class 3 (High)}	& \textbf{Recall (%)}
\textbf{Class 0 (Normal)}	& \textbf{2,010}	& 4	& 1		
& 995	& 66.78%	\			
\textbf{Class 1 (Micro)}	& 1	& \textbf{0}	& 0	&	
0	& 0.00%	\			
\textbf{Class 2 (Intermediate)}	& 1	& 0	& \textbf{1}		
& 0	& 50.00%	\			
\textbf{Class 3 (High)}	& 68	& 0	& 0	&	
\textbf{78}	& 53.42%	\			
\end{tabular}					
\end{table}					

Analysis of Classification Errors

1. *Majority-Class Bias and Overprediction of Class 3*:

The model correctly classified 2,010 normal sinus rhythm segments (Class 0) but misclassified 995 normal segments as Class 3 (High Burden). This high rate of false positives (33.05%) is a direct side-effect of the random oversampling (ROS) applied to the training

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split, which forced the model to sensitize itself to AFib characteristics (like irregular R-R intervals) to combat the 95.38% native normal class imbalance.

2. *Minority Class Detection Under Extreme Scarcity*:

Class 1 (Micro-Burden)* had only 3 total instances in the local database cache, with only 1 entering the test split. The model misclassified this single instance as Class 0.

Class 2 (Intermediate Burden)* had only 4 total instances, with 2 entering the test split. The model successfully identified 1 of the 2 instances (50% Recall), demonstrating that the spatial-temporal network can capture active paroxysmal fluctuations even with extremely low representation.

3. *High Burden Classification Performance*:

For *Class 3 (High Burden / Persistent AFib)*, the model achieved a Recall of 53.42% (78 out of 146 actual cases). The 68 cases misclassified as Class 0 (Normal) represent false negatives, where the network failed to detect subtle AFib traits within the 2-second input window. This highlights the clinical limitation of short 2.0-second snapshots for patients whose persistent AFib might temporarily manifest as regular rhythms or regularized pacing.

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Chapter 5

CONCLUSION

This study successfully developed and evaluated a hybrid 1D CNN-LSTM network to automate the classification of atrial fibrillation burden tiers from single-lead ECG signals. The architecture addresses the diagnostic limitations of standalone networks and alternative recurrent models by simultaneously capturing localized waveform morphology and long-range sequence intervals.

Experimental benchmarking demonstrates that the proposed 1D CNN-LSTM network establishes a highly effective diagnostic pipeline, achieving a dominant Recall rate of 42.55% and an ROC-AUC score of 0.6901 when evaluated against a raw, clinically skewed testing cohort. While traditional models like the 1D CNN-GRU exhibit inflated raw accuracy scores by defaulting to majority-class normal predictions, the proposed model maintains stable sensitivity across minority-burden tiers, minimizing the risk of false negatives that compromise patient safety in continuous monitoring environments.

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The technical validity of the model's forward path is confirmed through algorithmic cross-validation, where deep learning classifications directly correlate with traditional heart rate variability metrics, specifically the Root Mean Square of Successive Differences (RMSSD). Furthermore, visual verification via 1D Grad-CAM backpropagation demonstrates that the network isolates true physiological landmarks, specifically irregular R-R interval boundaries and chaotic fibrillation noise, rather than arbitrary background sensor noise.

The implementation of a low-latency FastAPI backend integrated with a SQLite database layer for historical scans, a rule-based CHA2DS2-VASc stroke risk calculator, and a responsive React user interface establishes a viable technical framework for converting deep learning models into accessible clinical decision-support tools. Future work will focus on expanding dataset diversity to incorporate broader patient demographics, evaluating alternative deep learning topologies, and testing the software pipeline within live, real-time clinical screening workflows.

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